Adaptive Hash Attention and Lower Triangular Network for Hyperspectral Image Classification

Zixian Ge, Guo Cao, Youqiang Zhang, *Member, IEEE*, Xuesong Li, Hao Shi, and Peng Fu

***Abstract*— Convolutional neural networks (CNNs), a kind of feedforward neural network with a deep structure, are one of the representative methods in hyperspectral image (HSI) classification. However, redundant information and interclass interference are common and challenging problems in HSI classification. In addition, if the spectral and spatial information is not properly extracted and analyzed, it will affect the classification performance of the network to a great extent. Aiming at these issues, this article proposes an HSI classification method based on an adaptive hash attention mechanism and a lower triangular network (AHA-LT). First, the attention mechanism is introduced in the preprocessing stage, which is composed of the spectral attention module and the adaptive hash spatial attention module in series. Then, the data processed by the attention mechanism are introduced into the lower triangular network (LTNet) to obtain the fused high-dimensional semantic features. Finally, we compress the features and obtain the output classification results through several fully connected layers. Among them, LTNet is composed of 2-D–3-D CNN and multiscale features. The network integrates the characteristics of multibranch, feature fusion, feature compression, and skip connections. Extensive experiments on four widely used HSI data sets show that the proposed method can obtain a great improvement in performance compared with the existing methods.**

***Index Terms*— 2-D convolutional neural network (CNN), 3-D CNN, deep learning, feature fusion, hash encoding, hyperspectral image (HSI) classification, spectral–spatial attention mechanism, skip connection.**

I. Introduction

HYPERSPECTRAL remote sensing is used to obtain the spectral and spatial data of ground objects by detecting electromagnetic waves reflected by hyperspectral sensors. This makes it easier to detect targets that cannot be detected by wideband remote sensing. Therefore, the emergence of hyperspectral remote sensing has great significance. Differing from traditional natural images, a hyperspectral image (HSI) is a very detailed segmentation of the image in the spectral dimension, which contains hundreds of channels. HSI data look like a data cube, containing one spectral dimension and two spatial dimensions. Thus, HSIs provide more detailed information on the ground for various practical cases [1], [2].

Due to the special properties of HSIs, in recent years, HSI research has become a hotspot in remote sensing. Among them, the HSI classification task plays an important role in mineral exploration [3], [4], army research [5], and precision agriculture [6], [7], and has been highly studied by researchers. The main goal of HSI classification is to classify the ground objects in an image at the pixel level. We usually use prior information in the image, such as a small number of training samples, to learn to distinguish the corresponding ground feature categories of other pixels in the HSI.

In recent decades, deep learning technology has become a potential feature extraction method in industry and academia. Compared with traditional methods, its performance has been greatly improved, and it shows strong advantages in HSI classification tasks. There are many deep learning-based models, such as a stacked autoencoder (SAE) [8]–[11], a restricted Boltzmann machine (RBM) [12], a convolution neural network (CNN) [13]–[16], a recurrent neural network (RNN) [17]–[19], a generative adversarial network (GAN) [20], [21], and a graph convolution network (GCN) [22]–[24].

CNN significantly improves the performance of HSI classification tasks due to its local perception and parameter sharing. Through the fixed shape of the convolution kernel moving in the data and position-by-position convolution, the CNN extracts different location features. Therefore, a large number of HSI classification tasks are CNN-based models [25]–[28]. In [29], an end-to-end dense spectral–spatial convolution (FDSSC) framework is proposed, which extracts the spectral and spatial features of HSI through different convolution kernel sizes. Guo and Zhu [30] designed a spatial feature fusion algorithm (CSFF) for HSI classification. Based on center loss, the algorithm can extract spectral and spatial features with good discriminative. The classification results are then obtained by fusing the two features. Li *et al.* [31] proposed a two-stream 2-D CNN structure. This method separately extracts global features and local features by the two-stream network for classification. In [32], a lightweight network model was proposed, which uses hybrid 2-D–3-D CNN for feature extraction. Gong *et al.* [25] combined

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multiscale CNN (MS-CNN) to design a classification network. The MS-CNN further enriches the features, which obtains good classification performance. Chen *et al.* [33] proposed automatic CNN models that can adaptively select parameters, avoiding the problem of manually adjusting parameters for different data sets. Hamida *et al.* [34] designed a 3-D deep architecture that simultaneously extract spectral and spatial features to obtain accurate classification results. He *et al.* [35] proposed a handcrafted feature extraction method. Combined with covariance maps and multiscale strategy, the extracted features have good discrimination. A 3-D-CapsNet architecture was given in [36]. This method addresses the noise and outlier problem according to the maximum correntropy criterion, and has good robustness and generalization ability. Hang *et al.* [37] designed a hyperspectral and LiDAR data classification model with a coupled CNN. Zhong *et al*. [38] proposed a residual network based on spectral and spatial features for HSI classification. The convolutional layers are connected through the residual block, which is more conducive to backpropagation. Ge *et al*. [39] combined 2-D CNN and 3-D CNN to construct a multibranch neural network for HSI classification.

To obtain a better classification results, strong representative and good discriminative features are essential. Due to the particularity of HSIs, hundreds of bands could be mixed with noise or redundant information, with some of the pixels in the image cube not belonging to the same category. In addition, different spectral bands and spatial information play different roles in extracting features. The usual CNN-based models have difficulty in solving the above problems. Therefore, scholars put forward the idea of an attention mechanism, that is, giving different weights to the data with different contribution rates, taking the essence, and discarding the dregs. The attention module learns and extracts the weight distribution from the features and applies the weight distribution to the original features in order to change the distribution of the original features, thereby enhancing the effective features and weakening ineffective or less contributive information. Squeeze-and-excitation networks (SENets) [40] are an early form of the attention mechanism. However, SENet only focuses on layers that have a stronger feedback ability at the channel level, and does not reflect attention in the spatial dimension. In 2018, Woo *et al.* [41] optimized SENet and proposed a convolutional block attention module, which applies attention to both the spectral and spatial dimensions. Similar to SENet, this module can be embedded in most mainstream networks and improves the feature extraction ability without significantly increasing the amount of computation or the number of parameters. The introduction of the attention module not only tells the network model what to pay attention to, but also enhances the representation of specific areas. To date, deep network models based on attention for preprocessing and postprocessing have come one after another, with all achieving certain results.

In terms of the preprocessing-based attention model [46], the original HSI cube is first processed by attention, and then the processed data are passed into the neural network. Mou and Zhu [42] designed spectral gates according to global convolution that adaptively emphasize important information and weaken unimportant content. In [43], a cooperative spectral–spatial attention dense network was proposed to adaptively extract important features of the images. This method generates an attention cube through spectral and spatial attention mechanisms. The attention cube can directly assign corresponding weight to each pixel of the original image cube. In [44], a mixed CNN with covariance pooling for HSI classification was proposed. In this method, channelwise shift and channelwise weighting are designed to adjust the order and weight of spectral bands. Another spectral–spatial attention network was proposed in [45]. In this model, the original image cubes were first studied by spectral attention and spatial attention and then input to the spectral–spatial feature learning module to obtain the result label. In [46], two network branches are designed to separately extract the spectral and spatial features. The two branches add the attention module in the preprocessing stage and then classify by fusing the two features.

Another candidate structure is to perform spectral and spatial attention processing on high-level semantic features, that is, a postprocessing-based attention mechanism [47], [48]. Ma *et al.* [49] proposed a double-branch multiattention mechanism network (DBMA), which designs two branches for spectral and spatial features extraction. Two types of attention modules are applied in these branches to ensure that more discriminative features are extracted. Based on the network structure of DBMA, Li *et al.* [50] adjusted the attention model to further improve the classification performance. An attention multibranch CNN model combined with an adaptive region search mechanism was proposed in [51]. This method designs a multibranch attention mechanism to enhance the discriminative branches and suppress less useful ones. Haut *et al.* [52] introduced a classification method based on visual attention-driven technology, which uses a trunk branch attention module and a mask branch attention module to enhance the performance of the network. Tang *et al.* [53] proposed a 3-D octave convolution with a spatial–spectral attention network (3DOC-SSAN). This method extracts features through octave convolution and uses the postprocessing attention method to highlight the important data. In [54], an attention-aided CNN model was proposed. This model uses a spectral attention subnetwork branch and a spatial subnetwork branch for spectral and spatial features extraction.

Among the plenty of deep learning-based methods, CNN proved its powerful ability and great potential in the HSI classification [39]. However, there are still some problems in the CNN-based model. First, most of the existing models use a single branch to extract the hybrid spectral–spatial features or two branches to separately extract the spectral features and spatial features. The simple structure of the network model and the fixed input size of data may not be able to extract rich features. Second, most of the existing methods that build networks only depend on 2-D CNN or 3-D CNN alone. Using 2-D CNN alone cannot make full use of the information between adjacent bands of HSIs. The 3-D CNN is sensitive to noise and contains redundant information in the

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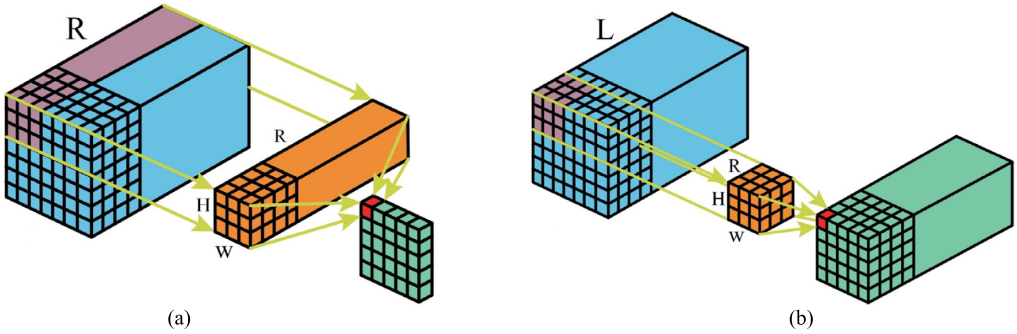


Fig. 1. The process of (a) 2-D CNN and (b) 3-D CNN.

image; using 3-D CNN alone still cannot extract the features well. Finally, different pieces of band information for HSIs have different contributions, and the neighboring pixels have different influences on the central pixel in the image cubes. It is often impossible to directly input unprocessed image cubes into the network to obtain a better classification performance. Dealing with the above problems is thus our motivation for this article. In the stage of data preprocessing, we use the spectral and spatial attention mechanisms to highlight the important contents of the data. Combining the ideas of multibranch networks, multiscale features, hybrid 2-D–3-D CNNs, and skip connections between branches, we design a feature extraction network. The shape of the network is similar to a lower triangular matrix, which in this article is called the lower triangular network (LTNet).

The main contributions of our work can be summarized as follows.

1) We introduce a network architecture based on an adaptive hash attention and a lower triangular network (AHA-LT) for HSI classification. Compared with the existing spectral–spatial-based networks, our model can not only effectively deal with the imbalance of information in data sets but also have the powerful ability to extract rich spectral and spatial features. Moreover, we use spectral padding and activation function Mish to further improve the performance.
2) The new spectral-wise attention mechanism and adaptive hash spatial-wise attention mechanism are designed in preprocessing. These attention mechanisms aim to selectively emphasize important features and suppress less useful features.
3) The proposed lower triangular deep neural network utilizes multiscale features, multibranch networks, skip connections between branches, and a hybrid 2-D–3-D CNN for feature extraction. The skip connections mechanism between branches can effectively realize feature reuse, which makes full use of low-level semantic features to obtain high-level spectral–spatial features with good discriminative and robustness.

The remainder of this article is organized as follows. Section II presents the related work. The detailed neural network model is introduced in Section III. Section IV describes the experimental data and results. Finally, Section V summarizes this article.

II. Related Work

A. Hash Encoding

Hash encoding is a one-way mapping relationship from a multimedia data set to a hash abstract set; that is, it uniquely maps a multimedia digital representation with the same perceptual content to a digital abstract [55].

In this article, the spectral sequence of each pixel in an image cube is mapped to a corresponding hash sequence. We construct the hash structure using the spectral vector $\boldsymbol{H}$ to generate the hash sequence $\boldsymbol{h}$. The construction formula is

$$h = \begin{cases} 1, & \text{if } H(i) > H(i-1) \\ 0, & \text{Otherwise} \end{cases} \tag{1}$$

where $\boldsymbol{h} = \{\mathbf{h}\{j\} | j = 1, 2, \ldots, m\}$ and i and m are the number of bands and pixels of the HSI cube, respectively.

B. 2-D CNN and 3-D CNN

At present, as one of the representative algorithms of deep learning, CNNs have achieved ideal results in large-scale image processing [56], [57]. Their weight sharing network structure likes a biological neural network, which can greatly reduce the complexity of the network model.

Two-dimensional CNNs and 3-D CNNs are common types of convolution layers in the field of HSI classification. Fig. 1 shows 2-D CNN and 3-D CNN diagrams for hyperspectral data, and their calculation formulas are as follows:

$$f_{i,j}^{x,y} = \Phi\left(\sum_{m}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} w_{ijm}^{hw} f_{(i-1)m}^{(x+h)(y+w)} + b_{ij}\right) \tag{2}$$

$$f_{i,j}^{x,y,z} = \Phi\left(\sum_{m}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\sum_{c=0}^{R-1} w_{ijm}^{hwc} f_{(i-1)m}^{(x+h)(y+w)(z+c)} + b_{ij}\right) \tag{3}$$

where $f_{i,j}^{x,y,z}$ represents the output value of position (x, y, z) in the jth feature map of the ith layer and Φ is the activation function. The size of the convolution kernel is (H, W, R), (h, w, c) represents the index of the convolution kernel, and m is the index of the feature map. In addition, w and b are weights and bias parameters, respectively, which are randomly initialized.

The 2-D CNN focuses on the spatial correlation of different pixels for a given image, and the convolution kernel moves in both the height and width of the data. A 3-D CNN is a generalization of a 2-D CNN. The convolution kernel depth of

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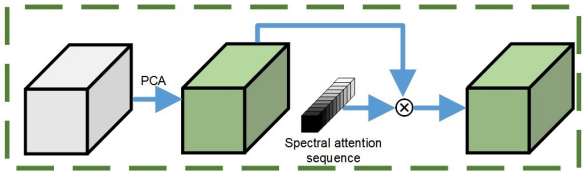


Fig. 2. Flowchart of spectral attention module.

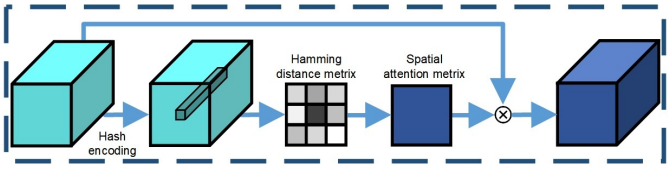


Fig. 3. Flowchart of spatial attention module.

a 3-D CNN is less than the input data channel size. Therefore, a 3-D CNN moves in three directions: height, width, and channel of the image. At each location, element-by-element multiplication and addition provide the output value of the 3-D CNN at the corresponding position. Due to the 3-D shape of the convolution core in the 3-D CNN, the output values are also arranged according to the 3-D shape. That is, the output is 3-D data. Compared with 2-D CNN, 3-D CNN also describes the spatial relationship of objects in 3-D space.

In HSIs, each pixel has hundreds of pieces of spectral band information, and only using 2-D convolution will lose the relevant information between the different bands of the image. The 3-D convolution extracts the spectral–spatial information of the image at the same time, which makes up for the shortcomings of 2-D convolution. However, if there is considerable redundant information and noise in the bands of the HSIs, using 3-D convolution alone cannot obtain good results. The effective combination of 2-D convolution and 3-D convolution can reduce the above defects, to some extent.

III. Attentional LTNet for HSI Classification

A. Spectral Attention Module

For an HSI processing, scholars verified that the redundancy from interband correlation in HSIs limits the classification performance, to some extent [39]. A large amount of band information will consume a significant amount of computing resources.

Recently, many HSI classification schemes have used PCA for reducing the dimensions of HSIs in data preprocessing stage [32], [39]. Its core involves calculating the similarity between different data, extracting the principal component according to the strength of the correlation, and completing the information fusion [58], [59]. PCA is a widely used method of dimension reduction because the data will not obviously lose useful information after dimension reduction [60]. In this article, PCA is used to preprocess the data of the input neural network. However, there is a problem that each band of the data cube processed by PCA has a different contribution to the overall data. When passing through a 3-D CNN with a fixed convolution core shape, different band information will be processed in the same way. Therefore, the important information in the image cube is not highlighted, which makes the extracted features less discriminative. Here, we want to highlight the important band information while weakening the bands that contribute less to the feature. This is the motivation of the spectral attention module proposed in this article. We combine the PCA and spectral weight attenuation coefficient s_1 to construct the spectral attention module. The module is shown in Fig. 2.

For the original HSI cube, we use PCA to reduce the dimensions in the spectral direction at first. After that, the bands of the image cubes are arranged in the order of variance from large to small. We place more emphasis on the bands with a large variance and weaken those with a small variance, therefore, setting an attenuation coefficient to construct the spectral attenuation sequence, Spe, in the following equation:

$$\text{Spe} = \left\{ s_1, s_1^{(2)}, s_1^{(3)}, \ldots, s_1^{(p)} \right\} \tag{4}$$

where s_1 is the spectral weight attenuation coefficient, and p equals to the number of spectral bands of the image cube after PCA. $s_1^{(p)}$ represents s_1 to the power of p. The value of s_1 is between 0 and 1, so the sequence Spe is arranged from large to small. Because the length of Spe is consistent with the number of bands of data cubes after PCA, we multiply the elements of Spe from large to small and the amount of band information of data cubes from large to small. Then, the reduced HSI data cubes are weighted by the Spe, that is, the weight of the spectral bands is gradually reduced according to the order of the band.

B. Spatial Attention Module

Each HSI cube contains much neighborhood information. If there are a large number of pixels with different label categories in the image cube (especially the image cube where the central pixel is at the category boundary of the original HSI), it will contain a significant amount of confusing information, which limits the learning ability of the network to a great extent, ultimately resulting in a low classification accuracy. In response to this problem, inspired by the technology of perceptual hashing image classification [61], we design a spatial attention module called adaptive hash attention. This module is located behind the spectral attention module, and the two modules are connected in series. The flowchart of the adaptive hash attention is shown in Fig. 3. The specific steps are shown as follows.

1) *Step 1:* For the image cube processed by the spatial attention module, we first convert each pixel of the image cube into a binary sequence containing only 0 and 1 through (1). Among them, for each pixel sequence $\boldsymbol{h} = \{H_1, H_2, \ldots, H_i\}$ in the image cube, if $H_i > H_{i-1}$, H_i is set to 1; otherwise, it is 0.
2) *Step 2:* The Hamming distance between the hash sequences of the neighborhood pixels and the central pixel is calculated. The formula for calculating the Hamming distance is as follows:

$$D'(h_1, h_2) = \frac{1}{m} \sum_{i=1}^{m} \left| h_1(i) - h_2(j) \right| \tag{5}$$

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where h_1 and h_2 are the hash sequences corresponding to two pixels and D' represents the normalized Hamming distance between h_1 and h_2.

3) *Step 3:* By synthesizing all of D', we can construct a matrix D'' with the same spatial size of image cube. Each element in D" corresponds to the hamming distance between the hash sequence of each neighborhood pixel in the image cube and the hash sequence of the center pixel.
4) *Step 4:* By setting the distance thresholds sequence $t = \{\tau, 2\tau, \ldots, (q-1)\tau\}$ from small to large, the elements in matrix D" are divided into q categories. The calculation formula of τ is as follows:

$$\tau = \left(\max\left(D''\right) - \min\left(D''\right)\right)/q \tag{6}$$

where q represents the distance category between the neighborhood pixels and the center pixel. The neighborhood pixels are divided into q categories by the thresholds. Similar to the spectral attention sequence, the spatial attention sequence is constructed by the spatial weight attenuation coefficient s_2. The length of the sequence is equal to q, Spa $= \{s_2, s_2^{(2)}, \ldots, s_2^{(q)}\}$.
5) *Step 5:* The center element of D" is set to 1, and the remaining elements are assigned the attenuation coefficient of *Spa* in descending order according to q categories in Step 4; thus, the spatial weight matrix D is constructed.
6) *Step 6:* Each pixel is given a corresponding weight by multiplying the weight matrix D with the image cube. At this point, each pixel in the corresponding image cube will be given corresponding weight at the spatial level according to its characteristics and then the corresponding spatial weight can be adaptively obtained.

For a better understanding, we use Fig. 4 to visualize the above steps. For the convenience of drawing, Fig. 4 only shows the spatial dimensions of the image cube, that is, each square represents a pixel. After mapping each pixel into a hash sequence, we first calculate the Hamming distance between the center pixel and all neighboring pixels, as shown in Fig. 4(a). Then, classifying the above distances according to the distance interval (here assume that there are five categories of distance intervals) to obtain (b). Among them, different colors indicate the Hamming distance of different distance intervals. Finally, the weight attenuation coefficients corresponding to different distance categories are assigned to the corresponding matrix positions to obtain the spatial weight matrix.

Through the above steps, we give a high weight to a pixel similar to the central pixel and a lower weight to the pixel far away from the central pixel.

C. Attention Modules and LTNet for HSI Classification

After processing of the two attention modules, the image cube is passed into the network. Our LTNet aims to extract robust and discriminative features for classification. The proposed network has five characteristics: hybrid 2-D–3-D CNN, multiscale features, multibranch network, skip connection, and feature fusion.

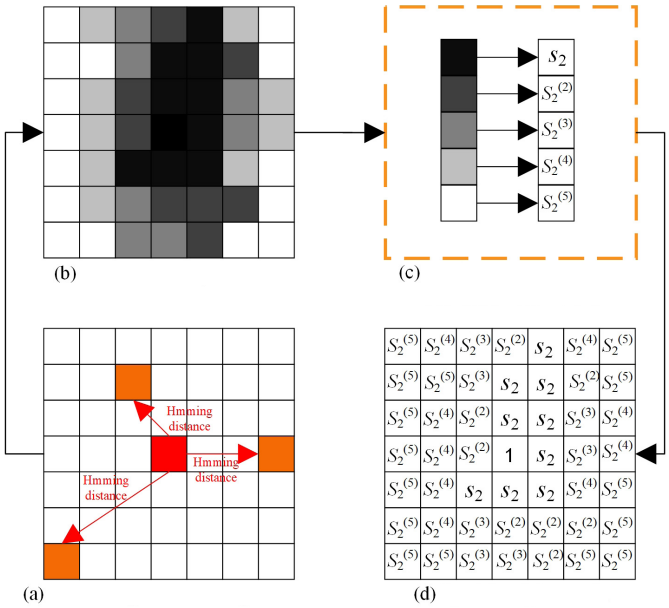


Fig. 4. Flowchart of spatial weight matrix construction. (a) Calculating the hamming distance. (b) Classification by distance. (c) Categories correspond to attenuation coefficient. (d) Spatial weight matrix.

Many existing neural networks are constructed only depending on 2-D CNN or 3-D CNN. The two-dimensional convolution kernel of 2-D CNN cannot learn the interband information, and only using a 3-D CNN is not effective in dealing with data redundancy. In addition, the convolution kernel of 3-D CNN is a 3-D cube, while 2-D CNN has a 2-D convolution kernel. The network using 3-D CNN alone needs more parameters to learn, and the training and testing time of the network will be longer. The use of 2-D–3-D CNN can improve the efficiency of the model. Therefore, the hybrid 2-D–3-D CNN gives full play to their advantages and weakens their disadvantages, to some extent. Multiscale strategy makes the features in the network more diverse, which improves the performance of the model to a certain extent. The proposed method uses different convolution kernels sizes in the first convolution layer of each branch, so that the network can obtain different receptive fields, capture information on different scales, and greatly improve the performance of classification task.

Many recent HSI classification works use a single network branch to extract features. The extracted features have limitations, which result in an unsatisfactory classification performance. In general, increasing the depth and width of the network can improve the network performance. We know the sparse network structure of GoogLeNet [62], which produces dense data and not only increases the performance of a neural network but also ensures the efficiency of computing resources. In addition, the small-size convolution kernel in the network layer extracts every piece of detailed information, while the large-size convolution kernel covers most of the spatial scales and learns the domain features. Generally speaking, the low-level features have a high resolution, including more locations and details, but contain less semantic information and more noise. High-level features have stronger representative

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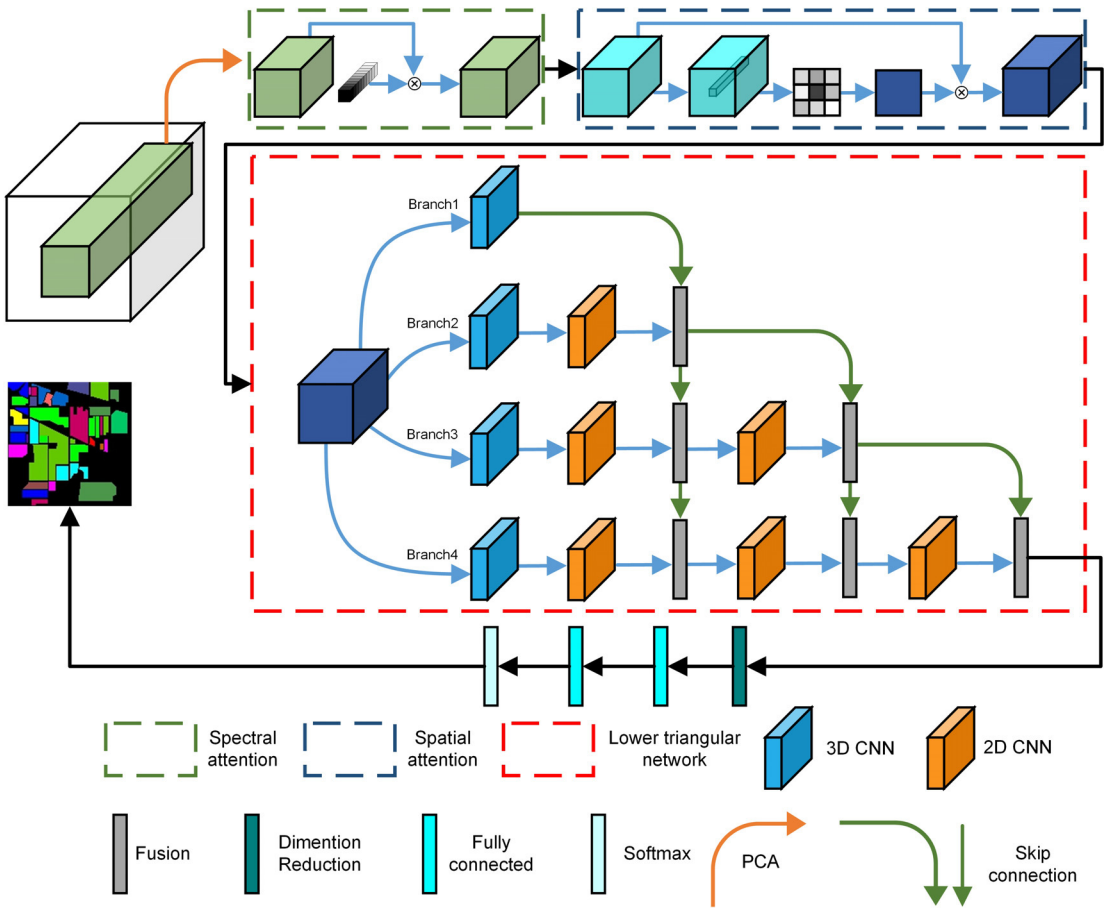


Fig. 5. Structure of the proposed method.

ability but a lower resolution and weak perception of details. In many tasks, integrating different scales of features is an important method for improving the classification results. Therefore, combining multibranch and multiscale strategy in the proposed LTNet can achieve a better performance.

There are two types of commonly used fusion methods, early fusion and late fusion. Early fusion generally exists in the front and middle part of the network. The fused features are used for further learning to obtain high-level semantic features. This type of method always uses concat or add. Late fusion enhances network performance according to fusing the results of different features. The early fusion method is adopted in this article.

CNN extracts the features of the low layer, middle layer, and high level of the image, and more network layers mean that richer features can be extracted. Moreover, the deeper the network layer is, the more abstract features are extracted. However, only increasing the network depth often leads to a gradient dispersion or gradient explosion. The emergence of ResNet [63] solves this problem. This network represents a layer as a residual function via a skip connection, making it easier to optimize the network and improving the network performance by increasing the depth. Inspired by ResNet, we designed a skip connection between the network branches. Therefore, further learning can be performed by connecting the output features of the layer in the corresponding location of each branch to the corresponding layer of the next branch. In addition, because the dimension of the fusion feature is high, we adopt a 3-D CNN layer to reduce it. Finally, the classification results are obtained by using two fully connected layers and a softmax layer. The neural network designed by the above methods has a strong feature extraction ability, and the extracted features can, to a great extent, highlight the characteristics of different categories of pixels in HSI. The proposed model is shown in Fig. 5.

The detailed structure and parameter settings of the proposed method are shown in Table I (taking the IP data set as an example). The spatial scale is 25×25, and all the spectral bands constitute the spectral–spatial information of a pixel. After preprocessing, the spatial scale remains unchanged and the spectral dimension changes to 30, 15, 15, and 15 for IP, PU, SA and BOT, respectively. At this time, the preprocessed image cubes are input into four parallel network branches. The network layers of the four branches are different and three of them are composed of hybrid 2-D–3-D CNN. The first layer of each branch is the 3-D CNN, and the other layers of the different branches are the 2-D CNN. Note that the size of the convolution kernel in different branches and different layers is different. The output feature of the first layer in each branch is fused with the output feature of the first layer in the next branch through a skip connection. Specifically, two feature maps are fused by concat operation in spectral direction. The second output feature of each branch (if it exists) and the second output feature of the next branch are fused by skip

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TABLE I
CONFIGURATION OF DEEP NETWORK USED IN FEATURE LEARNING PROCEDURE

Branch	Layer	Input shape	Output shape	Kernel-size	Filters	Connected to
1	Input	-	(25,25,30,1)	-	-	-
	Conv3d_1	(25,25,30,1)	(17,17,22,4)	9×9×9	4	Input
	Reshape_1	(17,17,22,4)	(17,17,88)	-	-	Conv3d_1
2	Input	-	(25,25,30,1)	-	-	-
	Conv3d_2	(25,25,30,1)	(19,19,24,8)	7×7×7	8	Input
	Reshape_2	(19,19,24,8)	(19,19,192)	-	-	Conv3d_2
	Conv2d_2	(19,19,192)	(17,17,64)	3×3	64	Reshape_2
	Fusion_2_1	((17,17,64),(17,17,88))	(17,17,152)	-	-	(Conv2d_2,Reshape_1)
3	Input	-	(25,25,30,1)	-	-	-
	Conv3d_3	(25,25,30,1)	(21,21,26,16)	5×5×5	16	Input
	Reshape_3	(21,21,26,16)	(21,21,416)	-	-	Conv3d_3
	Conv2d_3_1	(21,21,416)	(19,19,32)	3×3	32	Reshape_3
	Fusion_3_1	((19,19,32),(19,19,192))	(19,19,224)	-	-	(Conv2d_3_1,Reshape_2)
	Conv2d_3_2	(19,19,224)	(17,17,64)	3×3	64	Fusion_3_1
	Fusion_3_2	((17,17,64),(17,17,152))	(17,17,216)	-	-	(Conv2d_3_2,Fusion_2)
4	Input	-	(25,25,30,1)	-	-	-
	Conv3d_4	(25,25,30,1)	(23,23,28,8)	3×3×3	8	Input
	Reshape_4	(23,23,28,8)	(23,23,224)	-	-	Conv3d_4
	Conv2d_4_1	(23,23,224)	(21,21,16)	3×3	16	Reshape_4
	Fusion_4_1	((23,23,16)),(21,21,416))	(21,21,432)	-	-	(Conv2d_4_1,Reshape_3)
	Conv2d_4_2	(21,21,432)	(19,19,32)	3×3	32	Fusion_4_1
	Fusion_4_3	((19,19,32),(19,19,224))	(19,19,256)	-	-	Conv2d_4_2
	Conv2d_4_3	(19,19,256)	(17,17,64)	3×3	64	Fusion_4_3
	Fusion_4_4	((17,17,64),(17,17,216))	(17,17,280)	-	-	(Conv2d_4_3,Fusion_3_2)
Dimension reduction	Reshape_5	(17,17,280)	(17,17,280,1)	-	-	Fusion_4_4
	Conv3d_5	(17,17,280,1)	(11,11,180,1)	7×7×101	1	Reshape_5

connection. The feature obtained at this time contains rich information, but the dimension is high. Therefore, an extra 3-D CNN is used for a further feature extraction to achieve a dimension reduction. Finally, the final classification label is obtained through two fully connected layers and a softmax layer.

D. Measures Taken to Prevent Overfitting

Overfitting has always been a difficult problem in the training process of deep learning models. Too many feature dimensions, too many parameters, little training data, noise influence, and other factors may lead to a perfect prediction of the training set by the fitted function, while the performance is weak in the validation set and test set. That is, the prediction ability of the model for unknown samples is general, and the generalization ability is poor. In this article, the overfitting is improved by the synthesis of the following methods.

1) Padding Effect: For the 3-D CNN, as shown in Fig. 6, the data in the middle part are reused when the convolution core is moved, while the bands close to both sides are convoluted only once. When the HSI data are processed by PCA, the band of the data is arranged in order of variance from large to small, i.e., the previous bands contribute more to the data cube and contain more important information. This does not make enough use of the data, and the convolution processes will lose some information. Therefore, we adopt spectral padding processing [44] at both ends of the spectral direction of the image cube. The padding length and width are identical to the image spatial scale, and the depth depends on the size of the convolution kernel. Namely,

$$R = (r+1)/2 \tag{7}$$

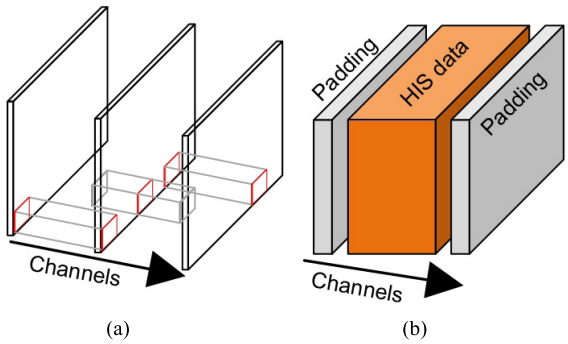


Fig. 6. Process of spectral padding. (a) Movement of the convolution kernel during the convolution process. (b) Way to add padding.

where R represents the depth of the padding cube and r is the depth of the convolution kernel.

After padding processing, each data point in the HSI image cube will be convolution at equal times, which effectively solves the above problem.

2) Strong and Appropriate Activation Function: Activation functions play a crucial role in neural network learning. The activation function adds nonlinear factors to the model so that the model can better fit various complex data. In the proposed model, Mish is used instead of ReLU. Mish is a smooth, continuous, self-regularized, nonmonotonic activation function [64]. Its definition is given in the following equation, and Fig. 7 visualizes the comparison between Mish and ReLU:

$$\text{Mish}(x) = x\tanh(\text{softplus}(x)) = x\tanh\left(\ln\left(1+e^{x}\right)\right) \tag{8}$$

where x represents the input of the function and ln $(\cdot)$ is a logarithmic calculation.

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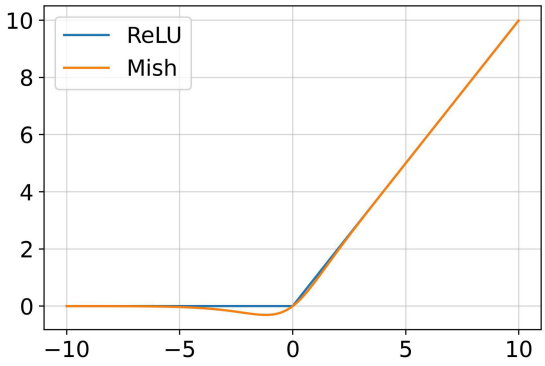


Fig. 7. Comparison of Mish and ReLU.

Compared with Mish, ReLU is a piecewise linear function, and it trims the negative values. That is, all network elements with negative inputs will no longer be activated. In this case, the network will loss some useful information. Mish preserves a small part of negative information, which improves the expressive ability of the model.

3) Dropout Layer: A dropout layer is a simple and effective method for preventing an overfitting phenomenon [65]. Through dropping out a percentage of units during forwarding propagation at the training stage, the synergy of feature detectors can be prevented to enhance the performance of the network.

There are many parameters in the deep learning model, and this article only uses 5% of the samples for training. The trained model easily produces an overfitting phenomenon. Therefore, in the training stage, a dropout strategy is adopted. We use a dropout with a 0.4 dropout rate for several fully connected layers at the end of the model. As a result, overfitting is avoided to a great extent, and the classification accuracy and generalization ability of the network model are improved.

IV. Experiments and Analysis

A. Data Description

Four HSI data sets were selected in our experiments: Indian Pines (IP), University of Pavia (PU), Salinas (SA), and Botswana (BOT). The three-band pseudocolor composite images and their corresponding reference maps are presented in Figs. 8–11.

The IP was obtained by an airborne visible/infrared imaging spectrometer (AVIRIS), 1992. The original data set contains 220 spectral bands with 145 × 145 spatial size, and the spatial resolution is 20 m per pixel. After discarding 20 water absorption bands, the remaining 200 bands ranging from 0.4 to 2.5 μm are used for the experiment. There contain a total of 16 categories.

The PU was collected by a reflective optics spectrographic imaging system sensor over an urban scene by Pavia University, Italy, 2001. The original data set contains 103 spectral bands ranging from 0.43 to 0.86 μm with 610 × 340 spatial size, and the spatial resolution is 1.3 m per pixel. The available ground truth for the PU contains nine categories.

The SA was collected by the AVIRIS sensor from SA Valley, CA, USA, 1998. The original data set contains 224 spectral bands ranging from 0.4 to 2.5 μm with 512 × 217 spatial size, and the spatial resolution is 3.7 m per pixel. There are a total of 16 categories.

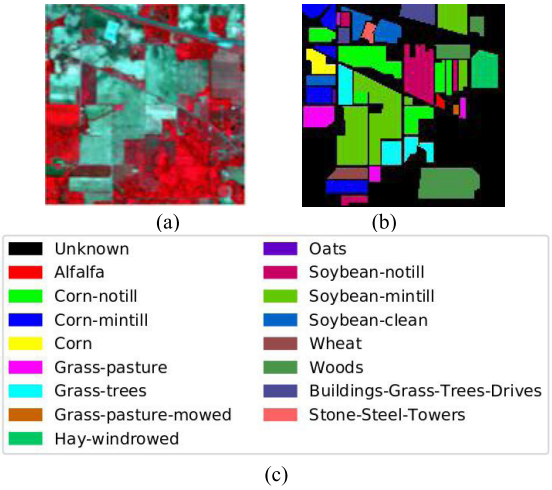


Fig. 8. IP data set. (a) False-color composite. (b) Ground reference map. (c) Land cover classes in the IP data set.

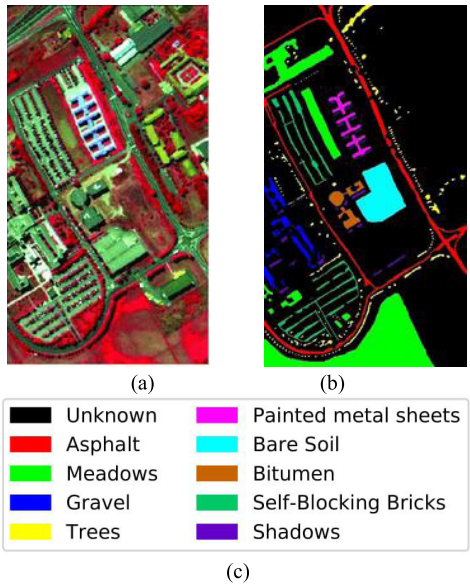


Fig. 9. PU data set. (a) False-color composite. (b) Ground reference map. (c) Land cover classes in the PU data set.

The BOT was acquired by the NASA Earth Observing-1 satellite over the Okavango Delta, BOT, on May 11, 2001. After removing uncalibrated and noise bands, there contain 145 bands ranging from 400 to 2500 nm with 1476 × 256 spatial size, and the spatial resolution is 30 m per pixel. There are a total of 14 categories in BOT.

B. Experimental Setup

All experiments are implemented on an Intel Core i7 3-GHz processor and a single NVIDIA RTX 2080Ti graphical processing unit (GPU). In the adaptive hash attention mechanism, the number of categories in the Hamming distance matrix corresponding to data sets IP, PU, SA, and BOT was set to 7, 5, 7, and 5, respectively. 0.0005 is chosen as the optimal learning rate, and the training epoch is set to 80 with a batch size 64.

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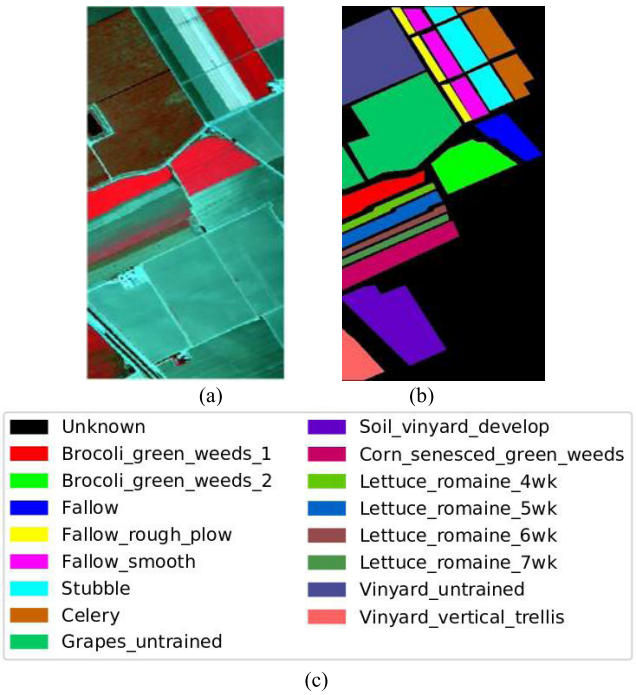


Fig. 10. SA data set. (a) False-color composite. (b) Ground reference map. (c) Land cover classes in the SA data set.

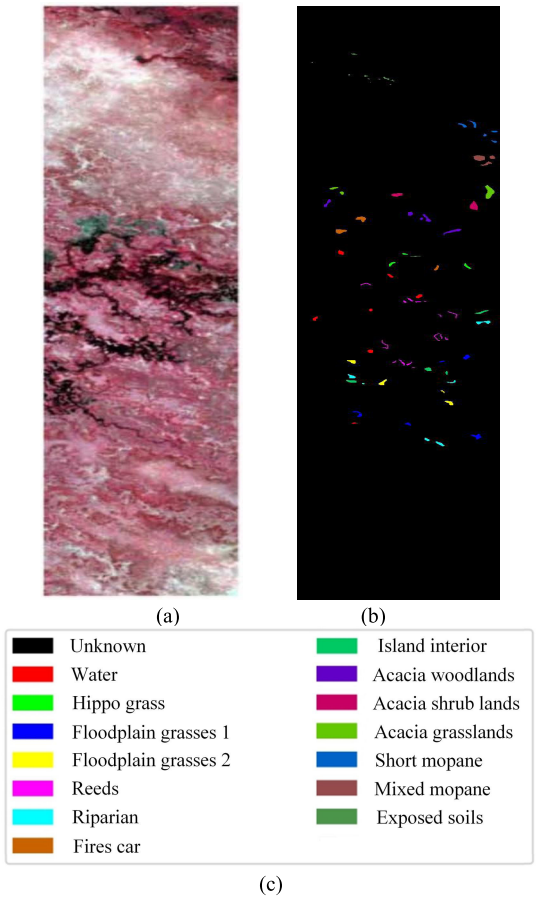


Fig. 11. BOT data set. (a) False-color composite. (b) Ground reference map. (c) Land cover classes in the BOT data set.

For all the data sets, we randomly select 5% of the data with labels for training, another 5% are used for validation, and the rest 90% are used as a test. The experiment is repeated ten times (in each experiment, the network parameters will be randomly initialized), and the average value and standard deviation of ten experiments are taken as the final result of the experiment. To analyze the results, three widely used quantitative metrics including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) were utilized to evaluate the classification performance.

The following are the methods included in our comparison.

1) SVM: the spectral information of pixels is extracted as features and classified by the SVM classifier.

2) 1-D-Auto CNN: The architecture of the 1-D-auto CNN is shown in [33], which uses the pixel vector of the HSI as an input. This method combines a 1-D CNN and a skip connection to construct the network. The input patch size is $1 \times 1 \times L$, where L represents the number of spectral bands.

3) 3-D-Auto CNN: The architecture of the 3-D-auto CNN is shown in [33], which chooses the neighborhoods of a pixel as an input. This method is based on a 3-D CNN and skip connection. The input patch size is $32 \times 32 \times L$.

4) SSRN: The architecture of the SSRN is shown in [38], which uses ResNet based on 3-D CNN to construct the network. The input patch size is $7 \times 7 \times L$.

5) FDSSC: The architecture of the FDSSC is shown in [29], which uses DenseNet to construct the network. The input patch size is $9 \times 9 \times L$.

6) HybridSN: The architecture of the HybridSN is shown in [32], which uses a 2-D–3-D CNN to combine a single network branch. The input patch size is $25 \times 25 \times L$.

7) SSAN: The architecture of the SSAN is shown in [46], which uses RNN and CNN to construct spectral and spatial feature extraction branches. The input patch size is $1 \times 1 \times L$ and $27 \times 27 \times L$ for spectral branch and spatial branch, separately.

8) DBMA: The architecture of the DBMA is shown in [49], which combines 3-D CNN, DenseNet, and attention mechanism in the network. The input patch size is $7 \times 7 \times L$.

9) CNN-MBFF: The architecture of the CNN-MBFF is shown in [39], which is based on a 2-D–3-D CNN, multibranch, and multiscale features. The input patch size is $25 \times 25 \times L$.

10) DBDA: The architecture of the DBDA is shown in [50], which combines 3-D CNN, DenseNet, and the attention mechanism in the network. Its architecture is similar to the DBMA, and the attention mechanism is different. The input patch size is $9 \times 9 \times L$.

The ten comparison methods include the classification methods using only spectral information as a feature, using the spectral–spatial feature, and using the method of introducing an attention mechanism based on the spectral–spatial information. In this way, the proposed method can be compared more comprehensively to prove the effectiveness at all levels. In addition, the network parameters and architecture for the four data sets are the same.

C. Classification Results and Discussion

1) Quantitative Evaluation: Tables II–V give information about the per-class accuracies, OAs, AAs, and Kappa obtained

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TABLE II
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE IP DATA SET

Class	SVM	1D-auto CNN	3D-auto CNN	SSRN	FDSSC	Hybrid SN	SSAN	DBMA	CNN-MBFF	DBDA	Proposed
1	10.87	37.74	75.61	86.67	96.51	85.29	93.48	75.93	82.00	100	100
2	66.74	76.61	90.30	96.99	87.84	96.69	87.46	95.89	93.84	98.46	97.90
3	45.90	64.18	79.86	99.11	95.96	83.45	94.12	93.22	92.06	95.89	97.40
4	8.02	51.86	90.23	97.00	96.26	94.48	100	100	100	99.51	99.09
5	69.15	95.87	82.10	99.01	99.28	81.02	96.19	95.48	93.24	96.73	97.58
6	90.68	89.57	94.85	98.33	98.65	89.18	96.82	99.22	94.67	94.36	98.66
7	17.86	6.98	100	86.96	84.29	100	100	69.57	100	80.00	100
8	50.63	100	96.63	96.01	98.33	100	98.90	100	100	100	100
9	30.00	10.00	90.00	66.67	80.77	85.00	100	76.92	100	100	100
10	62.35	91.56	96.31	82.55	89.94	92.61	92.50	93.64	98.98	93.68	96.26
11	83.30	84.36	90.04	90.89	97.79	94.44	93.79	91.56	97.59	99.03	97.09
12	23.78	84.43	95.10	92.02	97.49	92.93	95.14	92.84	98.22	97.03	99.01
13	79.51	86.78	95.45	100	100	95.45	98.98	100	95.78	100	99.40
14	90.28	90.07	97.53	94.87	96.39	99.83	93.40	95.05	97.31	96.60	99.32
15	12.18	54.43	93.13	95.24	93.98	91.82	98.22	95.45	98.64	95.89	98.88
16	7.53	90.00	98.68	98.82	97.08	89.19	95.65	97.67	96.47	98.81	85.57
OA(%)	65.94 ±2.77	78.89 ±2.03	91.34 ±1.17	93.56 ±0.30	94.81 ±0.51	93.20 ±0.51	93.74 ±1.24	94.46 ±0.35	96.34 ±0.27	97.19 ±0.21	**97.86 ±0.17**
AA(%)	46.80 ±5.26	69.65 ±3.66	91.61 ±0.99	92.57 ±0.57	94.41 ±0.71	91.96 ±0.44	95.91 ±0.92	92.03 ±0.22	96.18 ±0.24	96.62 ±0.26	**97.89 ±0.20**
Kappa(%)	59.63 ±5.03	75.99 ±2.85	90.08 ±1.08	92.65 ±0.43	94.10 ±0.40	92.24 ±0.53	92.85 ±0.99	93.67 ±0.41	95.83 ±0.18	96.80 ±0.33	**97.56 ±0.19**

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE PAVIA UNIVERSITY DATA SET

Class	SVM	1D-auto CNN	3D-auto CNN	SSRN	FDSSC	Hybrid SN	SSAN	DBMA	CNN-MBFF	DBDA	Proposed
1	91.49	68.64	93.01	99.31	99.77	98.37	98.97	99.28	99.73	99.87	99.59
2	97.73	97.86	98.55	99.77	99.89	99.92	99.87	99.77	99.86	99.65	99.90
3	70.75	64.15	92.92	97.83	99.78	95.69	96.12	97.57	99.50	99.91	99.01
4	88.67	81.60	98.20	99.85	99.89	98.79	98.25	98.95	98.62	98.58	98.76
5	98.59	88.17	99.84	99.97	99.92	100	100	99.53	100	99.86	100
6	86.38	93.19	99.58	99.64	99.97	99.86	99.85	99.90	99.83	99.95	100
7	82.41	78.53	83.92	96.67	99.84	98.25	99.76	99.78	98.44	99.94	99.59
8	90.90	67.22	82.30	97.34	94.32	97.89	96.68	98.33	98.72	96.38	98.76
9	99.26	78.55	87.65	99.65	100	90.22	95.84	97.04	99.42	98.91	100
OA(%)	92.45 ±1.71	85.82 ±1.99	95.28 ±2.06	99.28 ±0.22	99.37 ±0.21	98.93 ±0.36	99.07 ±0.29	99.34 ±0.23	99.58 ±0.14	99.38 ±0.13	**99.64 ±0.11**
AA(%)	89.58 ±2.39	81.64 ±1.81	92.89 ±2.67	98.89 ±0.27	99.26 ±0.38	97.67 ±0.40	98.37 ±0.23	98.91 ±0.24	99.35 ±0.09	99.23 ±0.19	**99.51 ±0.10**
Kappa(%)	89.94 ±2.66	81.15 ±2.14	93.73 ±2.34	99.05 ±0.34	99.16 ±0.41	98.59 ±0.39	98.77 ±0.35	99.13 ±0.18	99.45 ±0.17	99.15 ±0.16	**99.52 ±0.15**

by various methods on the four data sets. On the whole, the classification performance of the proposed method is better than that of the compared methods. Among the ten methods compared, the classification performances of the DBMA, CNN-MBFF, and DBDA are better, followed by the SSRN, SSAN, HybridSN, FDSSC, and 3-D-auto CNN. The classification performance of the 1-D-auto CNN and SVM is the weakest.

HSI is a type of special image that integrates spectral and spatial information. The 1-D-auto CNN and SVM extract the spectral information of pixels for classification. These methods ignore the spatial information and are easily disturbed by the phenomenon of "same material, different spectral" and "same spectral, different material" in HSIs. In IP, the OA of these two methods is less than 80%, while in the other three data sets, the classification accuracy is only approximately 90%. Therefore, the classification methods based on spectral information as features have poor performances.

Making full use of the spectral and spatial information of HSIs greatly improves the feature representation ability of the network, thus improving the classification performance. The 3-D-auto CNN is improved based on 1-D-auto CNN. It uses image cubes containing spectral and spatial information as the input on the network structure. The OA values of the four data sets increase by 12.45%, 9.46%, 6.41%, and 10.7%, respectively. Based on the idea of ResNet, SSRN utilizes a deep network branch to extract features and sets $7 \times 7 \times L$ image cubes as the input patch size of the network. The classification accuracy is higher than that of methods that only use spectral information. Combining the idea of DenseNet, FDSSC constructs a relatively deep network model. Compared with SVM and 1-D-auto CNN, its classification performance is also greatly improved.

The 3-D-auto CNN, SSRN, and FDSSC build the network architecture through 3-D CNN alone. However, when there is a considerable amount of redundant information

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE SALINAS DATA SET

Class	SVM	1D-auto CNN	3D-auto CNN	SSRN	FDSSC	Hybrid SN	SSAN	DBMA	CNN-MBFF	DBDA	Proposed
1	99.20	99.06	97.41	100	100	100	100	100	100	100	100
2	99.41	99.84	100	100	100	100	99.95	99.83	100	100	100
3	98.99	80.63	99.54	100	100	99.84	100	99.96	100	99.72	99.78
4	99.57	96.42	99.29	99.37	99.88	99.92	98.98	97.88	99.40	99.86	99.61
5	98.09	83.75	88.62	99.83	99.60	99.92	99.65	99.92	99.92	98.63	99.96
6	99.19	100	100	100	100	100	100	100	100	100	100
7	99.41	98.64	100	100	100	100	100	99.96	100	99.25	100
8	82.55	85.53	93.86	96.14	99.42	100	98.29	98.99	99.99	99.81	99.95
9	98.94	99.82	99.26	99.91	99.96	100	99.57	99.92	100	99.97	100
10	92.22	84.39	99.22	99.83	99.78	97.96	100	99.15	99.65	99.89	99.97
11	96.54	97.35	99.21	100	100	99.80	100	98.73	99.41	96.78	100
12	97.98	77.71	100	99.88	99.86	100	99.95	99.89	100	100	100
13	95.74	83.00	91.98	100	100	100	99.78	99.96	100	99.80	100
14	93.18	86.41	99.60	99.38	98.42	92.44	100	97.95	100	99.01	100
15	71.26	80.96	88.01	97.57	98.85	99.99	99.33	98.90	100	91.69	100
16	96.90	88.74	97.71	100	100	99.71	99.66	100	100	100	100
OA(%)	91.15 ±2.71	89.55 ±1.84	95.96 ±1.35	98.80 ±0.74	99.65 ±0.22	99.69 ±0.31	99.43 ±0.33	99.43 ±0.30	99.94 ±0.05	98.40 ±0.34	**99.97 ±0.02**
AA(%)	94.95 ±1.98	90.14 ±1.71	97.10 ±1.08	99.49 ±0.56	99.74 ±0.17	99.35 ±0.37	99.69 ±0.27	99.44 ±0.29	99.90 ±0.07	99.03 ±0.28	**99.95 ±0.03**
Kappa(%)	90.14 ±2.27	88.35 ±1.79	95.50 ±1.29	98.66 ±0.59	99.61 ±0.19	99.66 ±0.27	99.37 ±0.28	99.37 ±0.33	99.94 ±0.04	98.23 ±0.29	**99.96 ±0.02**

TABLE V
CLASSIFICATION RESULT OF DIFFERENT METHODS ON THE BOT DATA SET

Class Name	SVM	1D-auto CNN	3D-auto CNN	SSRN	FFDSSC	Hybrid SN	SSAN	DBMA	CNN-MBFF	DBDA	Proposed
1	99.20	77.40	98.47	99.33	99.60	97.34	97.31	99.19	96.95	97.62	100
2	99.41	88.37	100	98.18	100	100	100	99.63	100	98.40	100
3	98.99	100	94.29	100	99.37	92.21	98.67	100	100	100	100
4	99.57	100	98.56	95.93	99.00	99.49	95.75	97.19	96.67	97.69	100
5	98.09	73.70	89.55	94.07	87.91	97.66	98.80	91.34	98.76	94.11	98.80
6	99.19	97.46	100	97.25	95.60	100	94.72	99.44	94.76	99.16	98.80
7	99.41	84.44	95.74	100	100	100	99.19	100	100	100	100
8	82.55	43.86	100	95.34	100	95.08	100	100	99.48	98.92	100
9	98.94	100	86.21	99.43	100	88.96	99.66	100	100	100	100
10	92.22	75.54	88.72	99.43	99.78	100	93.83	99.42	91.05	97.51	96.61
11	96.54	96.31	100	100	100	85.55	97.96	99.88	100	100	100
12	97.98	96.00	99.01	100	99.11	96.95	100	100	90.15	100	89.78
13	95.74	94.81	79.50	100	100	92.96	87.27	98.83	84.81	100	100
14	93.18	100	95.45	100	100	100	100	98.65	100	100	100
OA(%)	91.15 ±3.06	82.59 ±2.91	93.29 ±1.99	98.46 ±0.71	98.18 ±0.14	95.14 ±0.49	96.84 ±0.41	98.73 ±0.19	96.34 ±0.16	98.76 ±0.21	**98.90 ±0.05**
AA(%)	94.95 ±2.71	87.71 ±3.05	94.68 ±2.36	98.50 ±0.82	98.60 ±0.20	96.16 ±0.66	97.37 ±0.39	98.83 ±0.29	96.62 ±0.19	98.82 ±0.17	**98.86 ±0.11**
Kappa(%)	90.14 ±3.35	81.12 ±2.83	92.73 ±2.27	98.33 ±0.62	98.45 ±0.16	94.73 ±0.51	96.57 ±0.50	98.62 ±0.25	96.03 ±0.08	98.65 ±0.25	**98.80 ±0.07**

and noise in the data set, using 3-D CNN alone cannot achieve a good classification performance. The 2-D CNN is less disturbed by redundant information than the 3-D CNN, so the hybrid 2-D–3-D CNN improves the classification performance, to some extent. HybridSN builds a lightweight network through hybrid use of 3-D CNN and 2-D CNN. It achieves OAs of 93.20%, 98.93%, 99.69%, and 95.14% in the four data sets. CNN-MBFF also uses a hybrid 2-D–3-D CNN to construct the network. Different from HybridSN, the CNN-MBFF network is composed of three branches. Each branch has different layers of CNN and different spatial sizes for the input data. The representation ability of the fusion features of the three branches is stronger. The classification accuracies of the four data sets are 3.14%, 0.65%, 0.25%, and 1.2% higher than that of HybridSN.

In recent years, scholars have conducted increasing in-depth research on images and proposed the attention mechanism. The attention mechanism has also been applied in the HSI processing field. SSAN adds the attention mechanism to its spectral feature extraction branch and spatial feature extraction branch, and obtains the classification accuracy of 93.74%, 99.07%, 99.43%, and 96.84% on the four data sets. DBMA and DBDA use two different attention mechanisms based on a similar network structure. With the help of the attention mechanism, the classification accuracy of the network model has been further improved. In the four data sets, the classification accuracies of DBMA and DBDA are higher than 94.4%, 99.3%, 98.4%, and 98.7%, respectively. Although CNN-MBFF does not use the attention mechanism, its OA is essentially higher than that of the method using the attention mechanism

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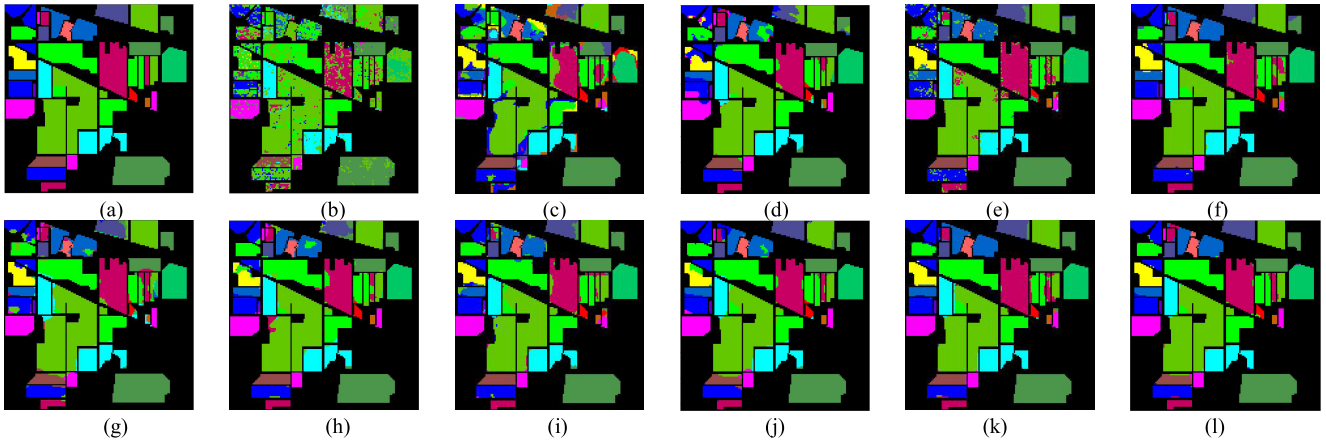


Fig. 12. Classification map of IP. (a) Ground truth. (b) SVM. (c) 1-D-auto CNN. (d) 3-D-auto CNN. (e) SSRN. (f) FDSSC. (g) HybirdSN. (h) SSAN. (i) DBMA. (j) CNN-MBFF. (k) DBDA. (l) Proposed method.

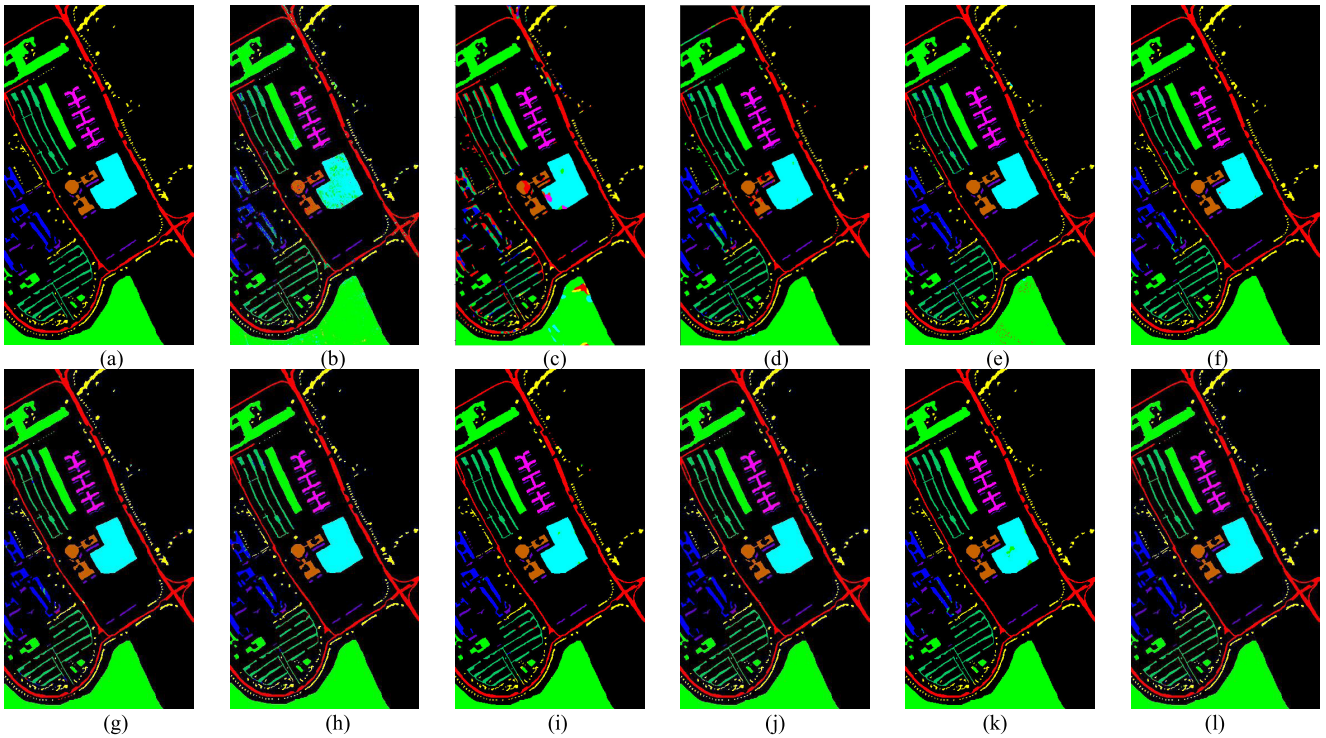


Fig. 13. Classification map of PU. (a) Ground truth. (b) SVM. (c) 1-D-auto CNN. (d) 3-D-auto CNN. (e) SSRN. (f) FDSSC. (g) HybirdSN. (h) SSAN. (i) DBMA. (j) CNN-MBFF. (k) DBDA. (l) Proposed method.

outside the BOT data set. It is true that the attention mechanism improves the network performance to some extent, but it belongs to the auxiliary mechanism. The classification performance mainly depends on the construction of the network model.

The proposed method constructs a powerful feature extraction network based on multibranch, multiscale features, hybrid 2-D–3-D CNN, interlayer skip connections, and feature fusion. In terms of the network structure, we use multiple network branches through composite connections between adjacent branches to form a more powerful feature extraction network. In this way, LTNet iteratively takes the output features of each layer of the previous branch as part of the input features and provides them to subsequent network branches in a stage-by-stage manner, finally using the feature map of the last network branch for HSI classification. The spectral and spatial attention mechanisms are used in the data preprocessing stage, which further improves the classification performance. Compared with the methods with the highest classification accuracy in the four data sets, the OA values of the proposed method can be improved by 0.67%, 0.06%, 0.03%, and 0.14%, respectively. From the classification accuracy data, we can see that the output feature of the proposed model has a strong representation ability and distinguishes different types of pixel features well.

2) Qualitative Evaluation: After quantitative analysis, we also display classification maps of all the compared methods. Figs. 12–15 visualize the ground-truth map and the classification results of compared methods on the four data sets. As seen in Figs. 12–15, the proposed method achieves

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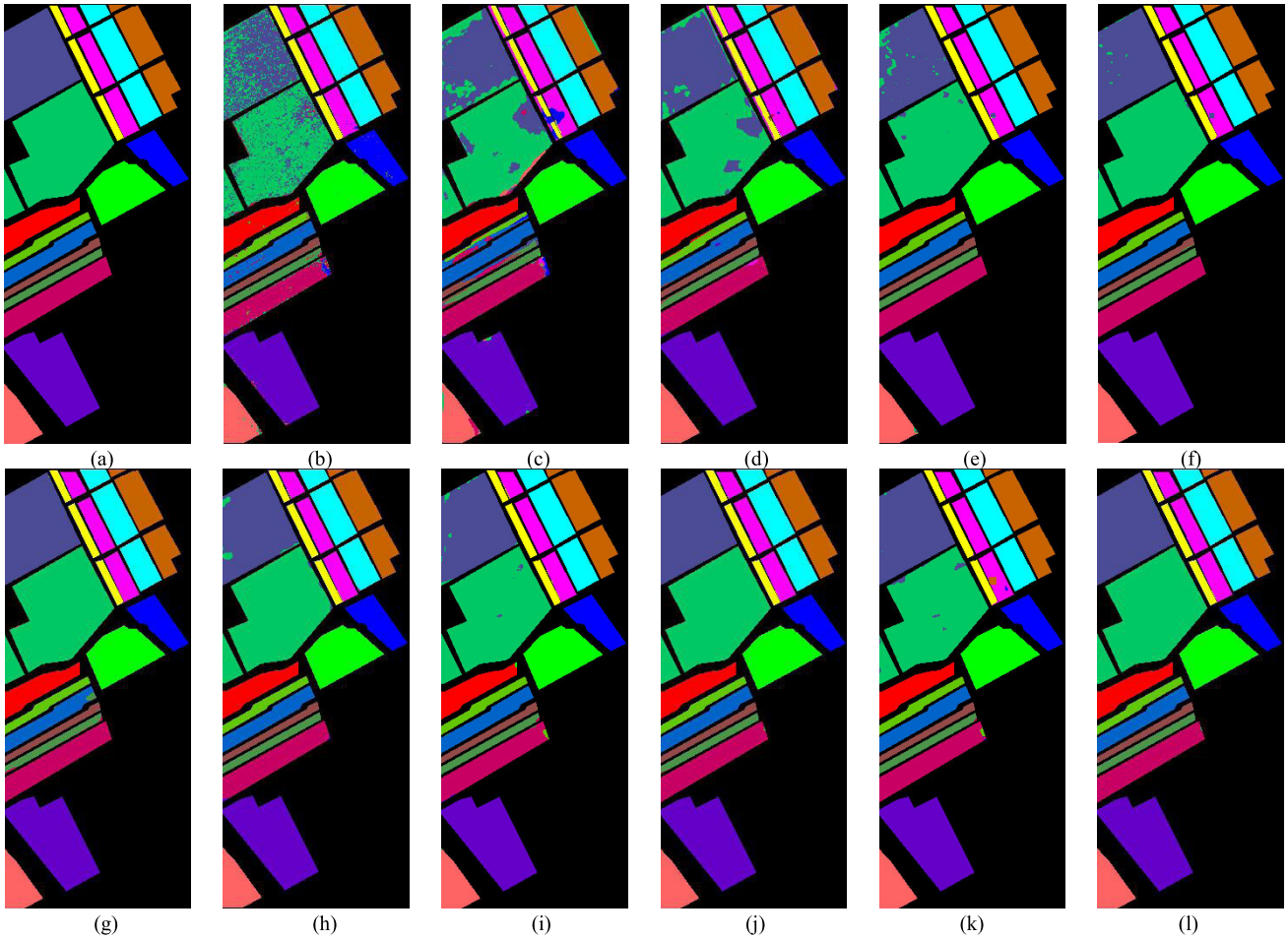


Fig. 14. Classification map of SA. (a) Ground truth. (b) SVM. (c) 1-D-auto CNN. (d) 3-D-auto CNN. (e) SSRN. (f) FDSSC. (g) HybirdSN. (h) SSAN. (i) DBMA. (j) CNN-MBFF. (k) DBDA. (l) Proposed method.

the best classification results in all compared data sets and obtains clean the classification maps. For SVM and 1-D-auto CNN, there are many noisy scatter points in their classification maps among four data sets [see (b), (c) in Figs. 12–15]. The 3-D-auto CNN, SSRN, FDSSC, HybridSN, SSAN, and CNN-MBFF optimize the network structure and enhance the feature extraction ability, which results in less noise in the classification maps. The network structure of SSAN is relatively simple, and the extracted features are weakly discriminative. Therefore, the noise in the classification results of SSAN is slightly higher. DBMA and DBDA are combined with an attention mechanism to improve the network performance, and the classification maps are relatively clean. But there are still misclassified pixels in some categories. Compared with the ground reference maps, the proposed network performs best in all compared data sets. The experimental results shows that the proposed method fully excavates the spectral and spatial features of HSIs through an attention mechanism and LTNet. The extracted features can well discriminate different categories of pixels.

3) Confusion Matrix Analysis: A confusion matrix, also known as the possibility table or error matrix, is a specific matrix used to present the visualization of algorithm performance. Each column represents the predicted value, and each row represents the actual category. Here, we only compare CNN-MBFF, DBMA, and DBDA with a better classification performance among the ten compared methods. The confusion matrices of different methods on the IP are shown in Fig. 16. The OA of the above three methods and the proposed method is high, but observing specific categories, there is still the phenomenon of a low classification accuracy. For example, in the case of CNN-MBFF, the Alfalfa pixels are mixed with Corn-notill pixels, and there are almost 18% mistaken pixels. There are also 6.16%, 7.94%, 6.76%, and 5.33% mistaken pixels in Corn-notill, Corn-mintill, Grass-pasture, and Grass-trees, respectively. In DBMA, 30.43% of Grass-pasture-mowed pixels are mixed with Alfalfa and Grass-pasture, 24.07% of Alfalfa pixels are mixed with Grass-pasture and Soybean-notill, and 23.08% pixels of Oats are mixed with Soybean-mintill and Soybean-clean. In addition, Alfalfa, Corn-mintill, Soybean-mintill, and Soybean-clean have 6%–9% mistaken pixels. DBDA has good classification performance, but 20% of Grass-pasture-mowed pixels are still mixed with Grass-pasture. Pixels in Grass-trees and Soybean-notill have more mistakes. In the proposed method, almost all the categories achieve classification accuracy higher than 96%, and only 14.43% of the Stone-Steel-Towers pixels are mixed with Soybean-clean, which indicates that the proposed method extracts discriminative and robust features to reach a high level of classification results.

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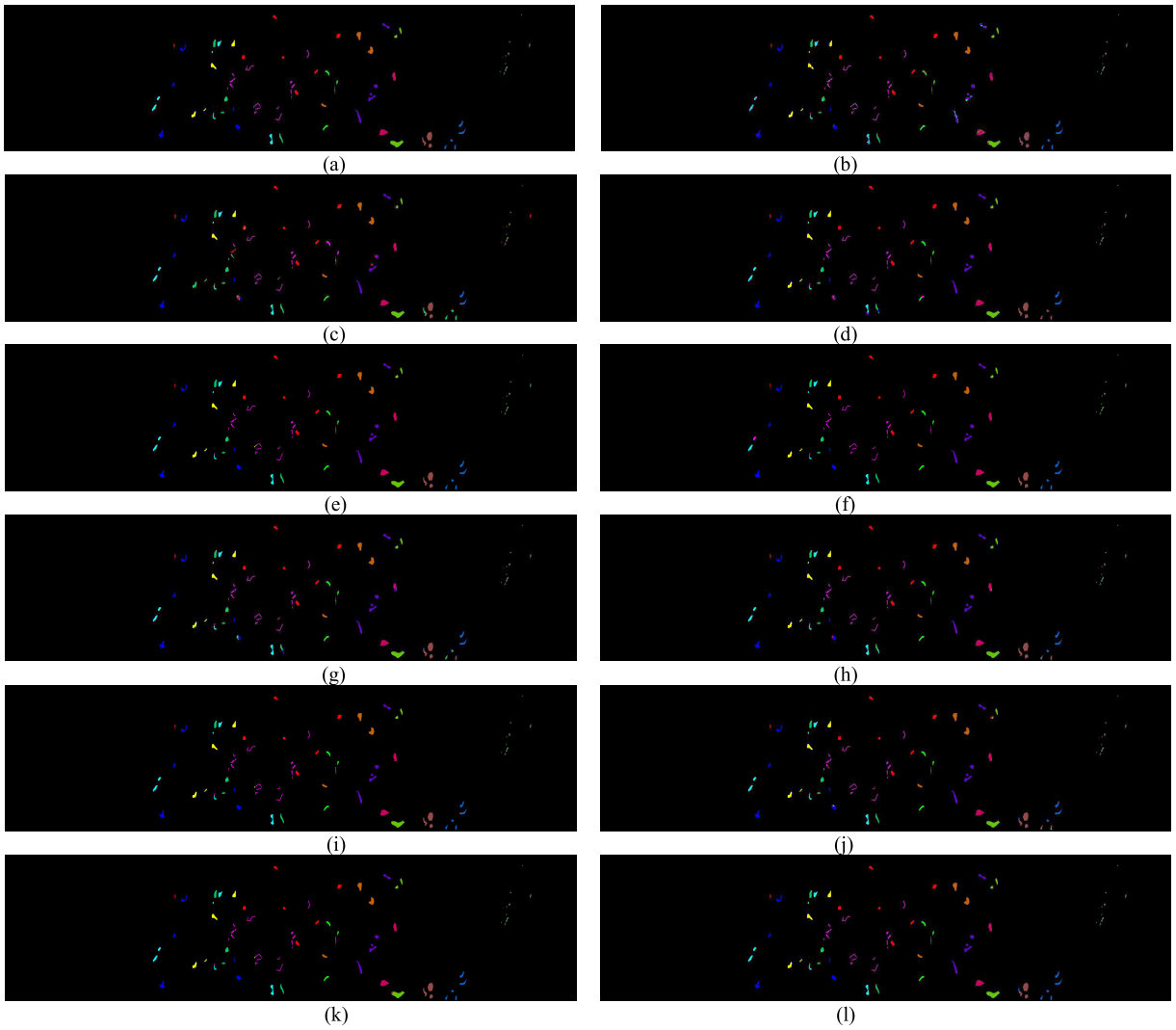


Fig. 15. Classification map of BOT. (a) Ground truth. (b) SVM. (c) 1-D-auto CNN. (d) 3-D-auto CNN. (e) SSRN. (f) FDSSC. (g) HybirdSN. (h) SSAN. (i) DBMA. (j) CNN-MBFF. (k) DBDA. (l) Proposed method.

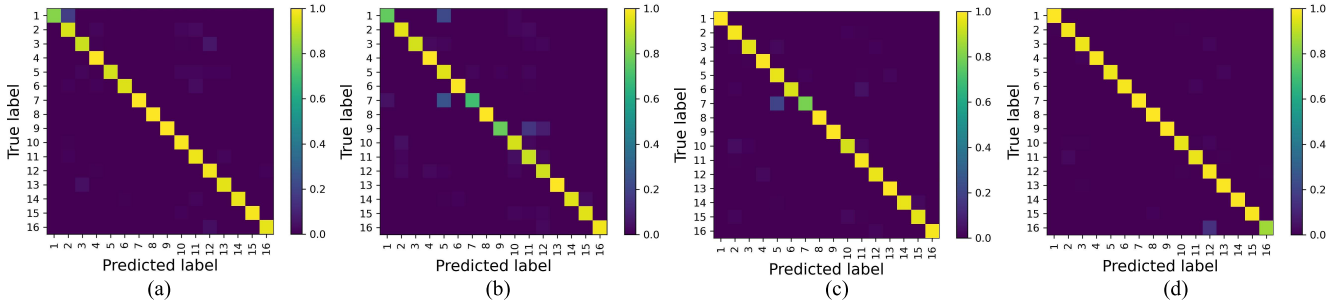


Fig. 16. Confusion matrix. (a) CNN-MBFF. (b) DBMA. (c) DBDA. (d) Proposed method.

4) Comparison of Different Methods When Different Training Samples Are Considered: To verify the generalization ability and robustness of the proposed method toward a different percentage of training samples, 3%, 5%, 8%, 10%, 15%, and 20% labeled samples were randomly considered as training data for the four data sets. The obtained results are shown in Fig. 17. When selecting 3% of the training samples, the proposed method achieves the best OA in PU and SA, approaches DBDA in IP, and reaches the middle level in BOT. With 5% of the samples for training, the proposed method can reach the highest OA in all data sets, indicating that the proposed method is able to better solve the overfitting phenomenon. As we feed more samples to the network, the proposed method still produces a high classification accuracy. In addition, 100% of the OA value can always be achieved with the fewest training samples in the PU, SA, and BOT. This indicates that

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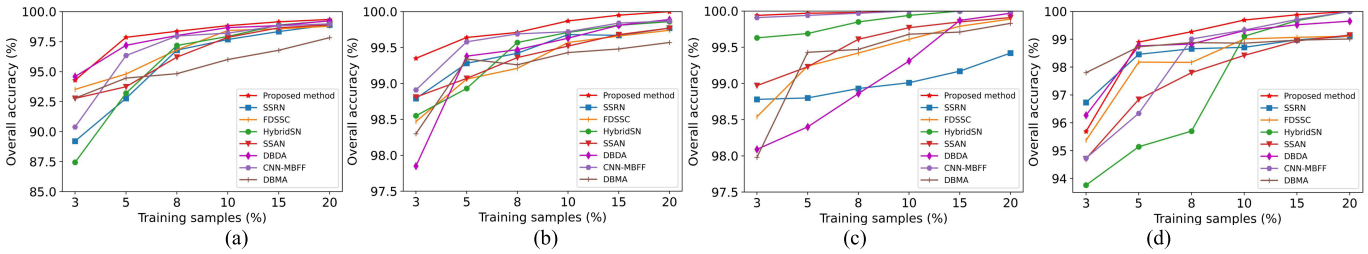


Fig. 17. OA results of the six methods with different numbers of training samples on (a) IP, (b) PU, (c) SA, and (d) BOT.

TABLE VI
STATISTICAL SIGNIFICANCE FROM THE MCNEMAR TEST ABOUT THE DIFFERENCE BETWEEN METHODS

Datasets	Methods	DBMA	CNN-MBFF	DBDA
IP		13.92	4.56	3.34
PU	Proposed	8.79	3.32	6.41
SA	AHA-LT	16.77	2.45	28.93
BOT		4.52	7.70	2.68

the feature extracted by the proposed method has a wealth of information, with a good discriminative, robustness, and generalization ability.

5) Analysis of Statistical Significant: the statistical significance of differences between the proposed AHA-LT and other methods was verified by McNemar test [66], which is defined by

$$Z = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}} \tag{9}$$

where f_{12} is the number of pixels that are classified correctly in method 1 but are incorrect in 2, while f_{21} is the number of pixels that are classified incorrectly in 1 but are correctly classified in 2. Z is the statistical significance value between the two methods. If Z between method 1 and 2 is larger than 1.96, the two methods are statistically significant at 95% confidence levels. $Z > 0$ indicates that method 1 is more accurate than 2; otherwise, $Z < 0$. As shown in Table VI, the McNemar test is all larger than 1.96. This confirms that the proposed method is superior to the abovementioned comparative methods.

D. Analysis of the Attention Mechanism's Effectiveness

In the proposed method, the original HSI data are processed by the spectral attention module and the adaptive hash spatial attention module, which make the data utilization more effective. The experimental results demonstrate that the network model with the attention mechanisms has a better classification accuracy. In this section, we discuss the difference in the OA of the proposed LTNet without an attention mechanism, using only spectral attention and then using only spatial attention. The classification results are displayed in Fig. 18.

As shown in Fig. 18, when the two attention mechanisms are removed, the classification accuracies of LTNet decrease by 1.73%, 0.89%, 1.23%, and 0.85% in the four data sets, respectively. After applying spectral attention and spatial attention, the accuracy of the network classification has been improved

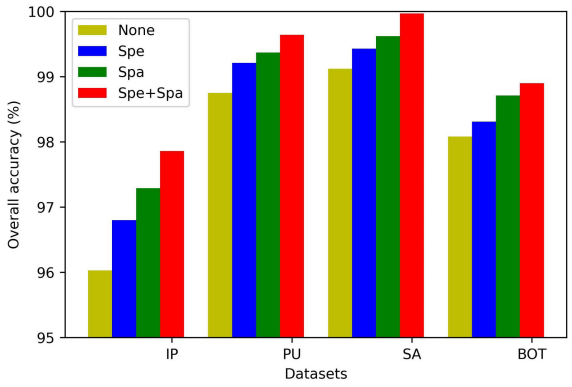


Fig. 18. Effectiveness of attention mechanisms.

to some extent, but it is still weaker than the combination of the two mechanisms. Although the classification accuracy of the network model is improved by combining the two attention mechanisms, the improvement of using only the spectral attention mechanism is smaller than that of using only the spatial attention mechanism. It can thus be judged that, for the four HSI data sets tested, the target pixels are more affected by neighboring pixels, so using the spatial attention mechanism improves the classification accuracy more obviously.

E. Attention Parameter Analysis

The two attention modules proposed in this article have coefficients: the spectral weight attention attenuation coefficient S_1 and the spatial weight attention attenuation coefficient S_2, respectively. Among them, S_1 is the spectral attenuation coefficient set between bands after PCA processing in the spectral attention module, and S_2 is the spatial attenuation coefficient set from near to far in different categories of distance from the central pixel in the adaptive hash spatial attention module.

We tested a total of 36 combinations of S_1 and S_2 with six coefficients from 0.75 to 1 and obtained the results shown in Fig. 19. In the process of S_1 and S_2 decreasing from 1 to 0.75, the classification accuracy first increases and then decreases. However, in most cases, the classification accuracy is higher than that when S_1 and S_2 are equal to 1, indicating that the spectral and spatial attention mechanisms improve the classification accuracy of the model. In Fig. 19, when S_1 is equal to 0.95 and S_2 is equal to 0.9, the classification accuracy of the model is the highest. These parameters are selected as the best parameters of the attention modules.

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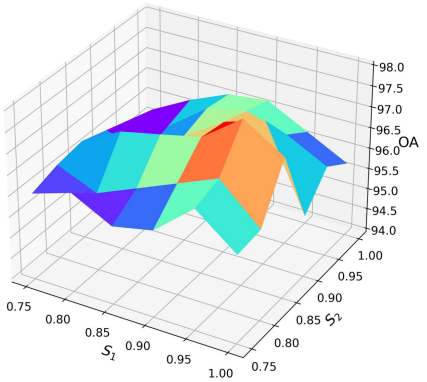


Fig. 19. Classification accuracy of IP data sets under different attention attenuation coefficients.

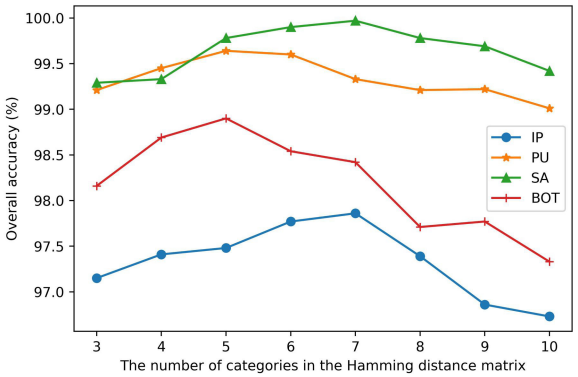


Fig. 20. Influence of the number of categories on classification performance in Hamming distance matrix.

F. Correlation Analysis Between the Number of Categories of Hamming Distance Matrix and Classification Performance

In the adaptive spatial attention mechanism, there is a parameter to be determined, that is, the number of categories of distance intervals q in the Hamming distance matrix. In four data sets, we set the Hamming distance interval categories from 3 to 10, a total of eight cases to test the classification performance of the model, and the results are shown in Fig. 20.

As can be seen from Fig. 20, among all data sets, when the number of categories of Hamming distances is very small, the classification performance of the model is low, which is close to the classification performance of using only spectral attention mechanism. With the increase of the number of categories, the classification performance of the model increases at first and then decreases. In other words, when the Hamming distance category is small, the effect of adaptive spatial attention mechanism is very weak. When the Hamming distance category is too large, some useful information in the data cube is excessively weakened, resulting in a decrease in classification performance. According to the optimal experimental results, we choose 7, 5, 7, and 5 as the category number of Hamming distance interval of IP, PU, SA, and BOT, respectively.

G. Comparison of the Influence of Different Numbers of Network Branches on the Classification Results

The third part introduces that the network model proposed in this article has four network branches and that the number of network layers in each network branch increases gradually. In this part, we explore the influence of different numbers of network branches on the classification results of the model. As the regulation of our proposed model, one, two, three, and five branches are separately selected to replace the proposed LTNet. The classification results performed on the four data sets are given in Fig. 21.

In Fig. 21, the classification performance of the network as a whole increases as the number of branches increases. However, when the number of branches exceeds 4, the classification of the network decreases slightly. This shows that, to some extent, by increasing the number of network branches, the model learns richer and higher level semantic information and thus obtains better classification results.

TABLE VII
COMPARISON OF OA OF ACTIVATION FUNCTIONS MISH AND RELU ON DIFFERENT DATA SETS

Method	IP	PU	SA	BOT
ReLU	96.02	98.33	98.41	97.11
Mish	97.86	99.64	99.97	98.90

H. Analysis of the Effectiveness of Padding and Dropout

To solve the overfitting problem and further enhance the classification performance of the proposed method, this article introduces band padding and dropout. After PCA processing, padding is performed on both sides of the data cube in the spectral direction, and dropout is used in the fully connected layers of the network. The experiment results are presented in Fig. 22.

In Fig. 22, we use A to denote padding and B to denote dropout. It can be seen that after using A and B alone, the classification accuracy has been improved. In addition, the joint use of A and B further improves the classification performance. The band padding and dropout used in this article bring performance improvements of 0.63%, 0.25%, 0.36%, and 0.31%, respectively, in the four data sets.

I. Comparison of the Influence of Mish and ReLU on OA

ReLU is a commonly used activation function, but this article uses Mish instead of ReLU. The OA of using two activation functions in LTNet is compared, as shown in Table VII.

The experimental results show that using Mish instead of ReLU does improve the network performance. In IP, PU, SA, and BOT, the accuracies increased by 1.84%, 1.31%, 1.56%, and 1.79%, respectively. As a nonmonotonic and smooth activation function, Mish proves its capability in enhancing the performance of neural network, which is worth deeper attention.

J. Training and Testing Time of Compared Methods Based on Attention Mechanism

Training time and testing time provide effective measures of the computational efficiency of the model. We compare the proposed method with SSAN, DBDA, and DBDM that use the attention mechanism, as shown in Table VIII.

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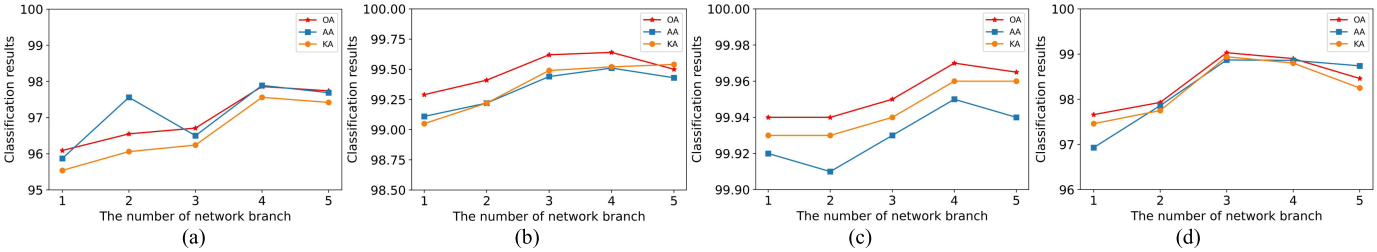


Fig. 21. Classification accuracy under different branches of proposed neural network. (a) IP. (b) PU. (c) SA. (d) BOT.

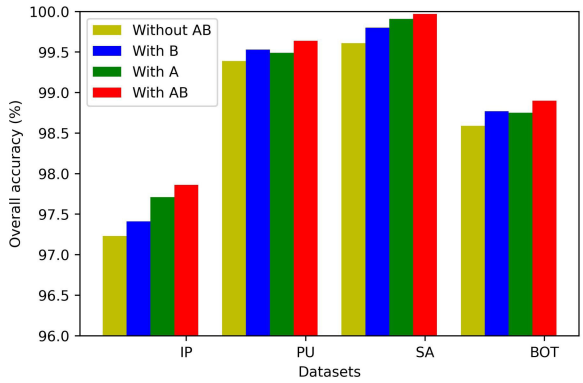


Fig. 22. Effective of padding and dropout.

TABLE VIII

CLASSIFICATION TRAINING AND TESTING TIMES OF DIFFERENT ATTENTION-BASED METHODS FOR FOUR HSI DATA SETS

		IN	PU	SA	BOT
SSAN	Train.(m)	4.2	16.67	24.86	2.03
	Test.(s)	9.34	55.89	105.33	6.67
DBMA	Train.(m)	2.90	12.27	20.57	1.71
	Test.(s)	6.37	47.64	87.34	4.65
DBDA	Train.(m)	1.81	11.61	25.01	1.62
	Test.(s)	5.58	42.78	99.53	5.58
AHA-LT	Train.(m)	5.29	13.82	17.77	1.16
	Test.(s)	13.51	28.55	37.43	2.53

It can be seen from Table VIII that the proposed AHA-LT model has the highest efficiency on both SA and BOT data sets. The training time of AHA-LT on PU is slightly longer than DBMA and DBDA, but the testing process is the fastest. For the IP data set, the efficiency of the AHA-LT is relatively low.

The structure of AHA-LT is simpler than DBMA and DBDA, and does not use a very deep network. In addition, AHA-LT converges very fast, and it can converge quickly within 80 epochs. Therefore, the overall efficiency of AHA-LT is higher. For IP, the spectral dimension of AHA-LT is compressed to be 30 after PCA (15 for other data sets), so the amount of data contained in image cubes is larger than that of other data sets, resulting in weaker computation efficiency.

V. CONCLUSION

In this article, we proposed a novel AHA-LT framework for HSI classification, which uses a multibranch CNN with skip connections as a baseline to extract the spectral–spatial features and combines an attention mechanism to select and enhance the spectral bands and spatial pixels. First, after the original HSI is reduced by PCA, the image patches apply different weights to different spectral bands through spectral attention. Then, based on the adaptive hash spatial attention module, different weights are given to the pixel fields in the image cube according to the Hamming distance from the central pixel. Finally, the image cube processed by the attention mechanism is passed into the LTNet for feature extraction. The experimental results showed that the proposed AHA-LT model achieves a state-of-the-art classification performance and is easily generalized to other HSI classification methods.

Our future work is to conduct further research and try to exploit a more efficient spectral–spatial joint attention mechanism; that is, a module is used to simultaneously complete the spectral attention and spatial attention mechanism. Furthermore, how to adaptively select the number of branches and network layers in a multibranch neural network instead of a manual setting is another research direction to study.

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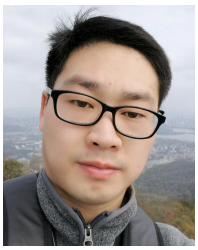
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